

CS425: Computer Networks

Homework-1 : 230316

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Figure 1

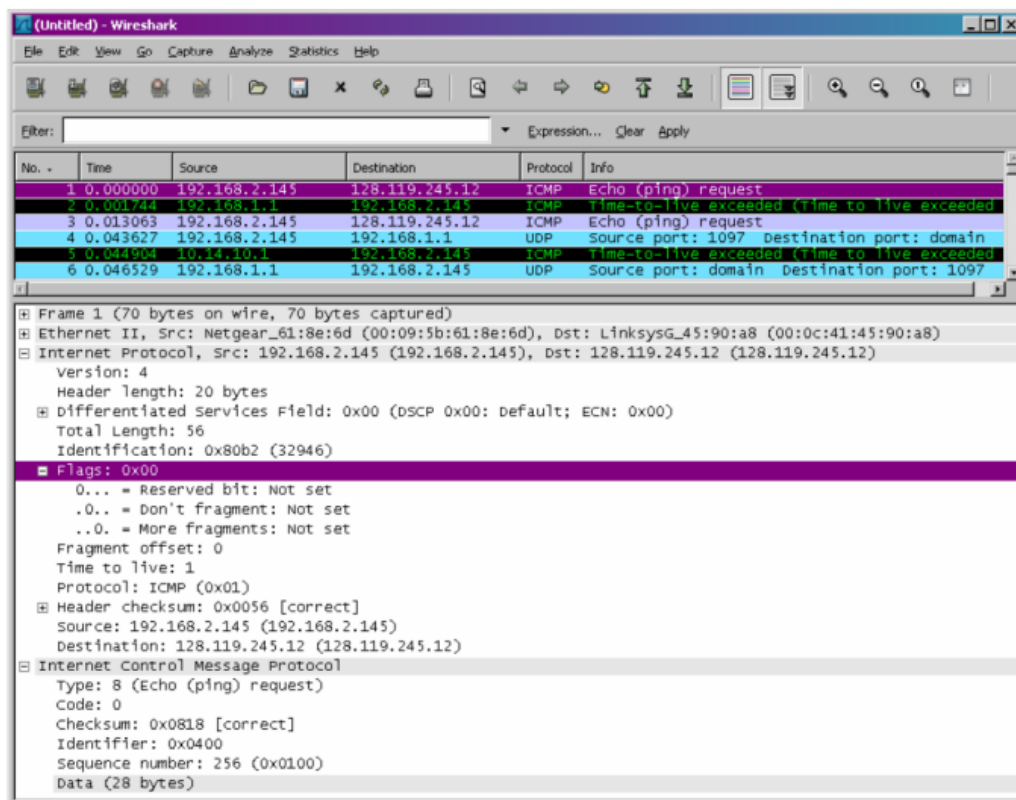


Figure 1: IPv4 packet header (Figure 1).

1. Upper Layer Protocol Field

From Fig. 1, the value in the Protocol field of the IPv4 header is:

0x01

According to the IPv4 protocol specification, the value 0x01 corresponds to ICMP (Internet Control Message Protocol). Therefore, the upper layer protocol carried inside this IP datagram is ICMP.

2. Header Length and Payload Size

From Fig. 1:

- Total Length = 56 bytes

- Header Length = 20 bytes

The Total Length field represents the size of the entire IP datagram, including both header and payload.

$$\text{Payload Size} = 56 - 20 = 36 \text{ bytes}$$

Hence, the IP datagram contains 36 bytes of data.

3. Has the Datagram Been Fragmented?

From Fig. 1:

- Fragment Offset = 0
- More Fragments (MF) flag = Not set

In IPv4 fragmentation:

- A non-zero fragment offset indicates a later fragment.
- The MF flag set to 1 indicates additional fragments follow.

Since the fragment offset is zero and the MF flag is not set, this packet represents a complete, unfragmented IP datagram.

4. Identification and TTL Fields

From Fig. 1:

- Identification = 0x80b2 (32946)
- Time To Live (TTL) = 1

The Identification field uniquely identifies fragments belonging to the same original datagram. The TTL field specifies the maximum number of hops the packet can traverse before being discarded.

Figure 2

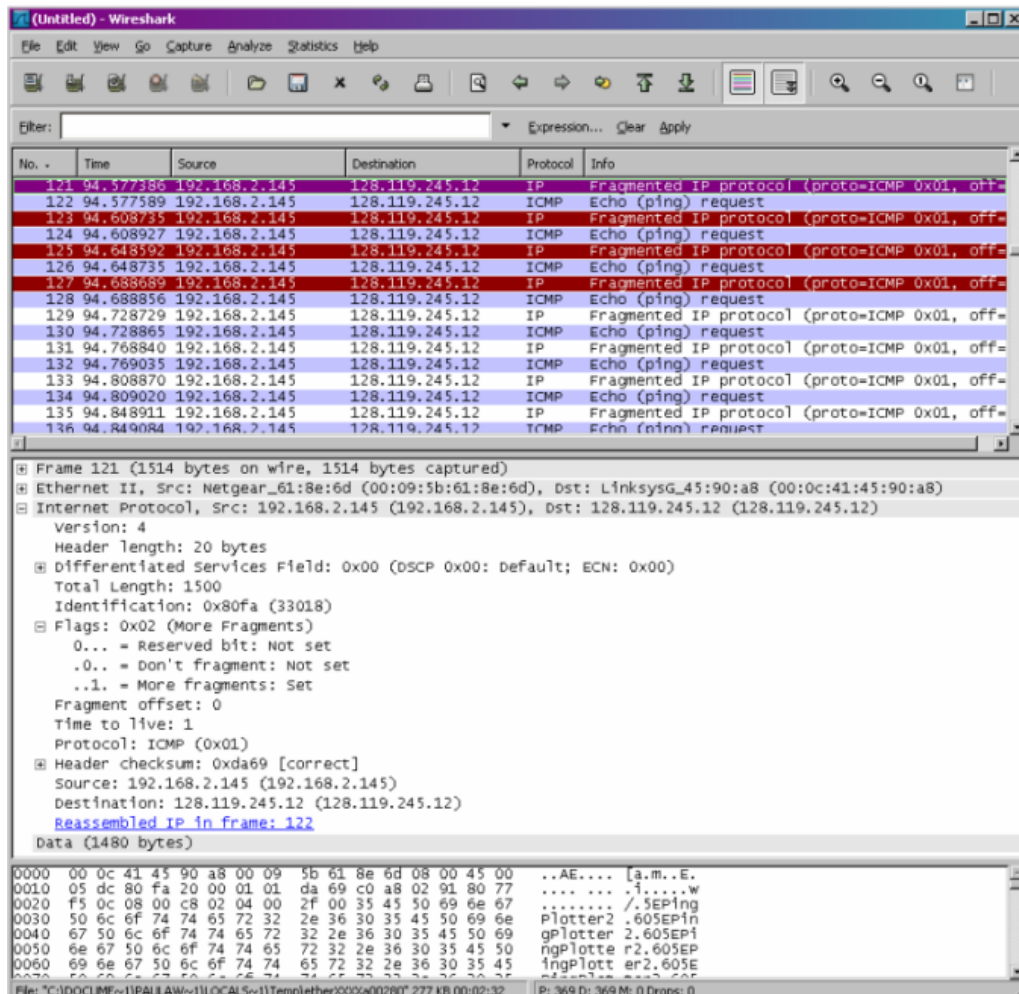


Figure 2: First fragment of IPv4 datagram (Figure 2).

5. Has the Message Been Fragmented?

From Fig. 2, the More Fragments (MF) flag is set.

When the MF flag is equal to 1, it indicates that additional fragments follow. Therefore, the original datagram has been divided into multiple fragments.

6. Which Field Indicates Fragmentation?

Fragmentation is indicated by the **Flags field**, specifically the More Fragments (MF) bit. If

$$MF = 1$$

then the datagram has been fragmented and more fragments are expected.

Thus, the presence of the MF bit set in Fig. 2 indicates fragmentation.

7. First Fragment or Later Fragment?

From Fig. 2:

9. Are There More Fragments?

From Fig. 3, the More Fragments (MF) flag is not set.

When the MF flag is 0, it indicates that no additional fragments follow.

Therefore, this is the **last fragment** of the original datagram.

10. Fields That Change Between First and Second Fragments

Comparing Fig. 2 and Fig. 3, the following IP header fields change:

- Total Length
- Flags (MF bit)
- Fragment Offset
- Header Checksum

From the figures:

- **Total Length:** 1500 in Fig. 2 and 520 in Fig. 3.
- **Flags:** 0x02 (MF set) in Fig. 2 and 0x00 (MF not set) in Fig. 3.
- **Fragment Offset:** 0 in Fig. 2 and 1480 in Fig. 3.
- **Header Checksum:** 0xda69 in Fig. 2 and 0xfd84 in Fig. 3.

Explanation:

- The **Total Length** differs because each fragment carries a different portion of the original datagram.
- The **Flags** field changes because the first fragment has the More Fragments (MF) bit set, while the last fragment has it cleared.
- The **Fragment Offset** changes to indicate the starting position of each fragment within the original datagram (measured in units of 8 bytes).
- The **Header Checksum** is recomputed for each fragment since header fields differ.

The Identification field remains the same in both figures, since it is used to associate fragments belonging to the same original datagram.